MATH 231 – Linear Algebra I

The Matrix Unit, Part 10 ▪ Cramer's Rule, the Inverse, and the Pseudo-inverse

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About this document. This is Part 10 of the matrix unit, Sections 50–54. Section numbers run continuously through the unit (Part 2's opening table lists them all).

Assumed. The determinant, its laws D1–D10 and cofactors (Part 7); the 2×2 determinant as an area and its column properties (§22–§23); the 2×2 inverse (§34); null space and rank–nullity (Part 6); elimination (Part 8); least squares and the normal equations (Part 9); orthogonal projections (§27).

What you should be able to do afterwards. Solve a small system by Cramer's rule and explain it with areas; state the equivalent conditions for an $n\times n$ matrix to be invertible; prove that a one-sided inverse of a square matrix is two-sided; compute an inverse by Gauss–Jordan elimination and by the adjugate; use the algebraic properties of the inverse; and compute and recognise the pseudo-inverse of a matrix with full column rank, with full row rank, or of rank one.

50. Cramer's rule

50.1. The statement

Theorem 50.1 (Cramer's rule)**.** *Let $\mathbf{A}\in\mathbb{R}^{n\times n}$ with $\det\mathbf{A}\neq 0$, and let $\mathbf{A}_i(\mathbf{b})$ denote $\mathbf{A}$ with its i-th column replaced by $\mathbf{b}$. Then the unique solution of $\mathbf{Ax}=\mathbf{b}$ is given, entry by entry, by*

$$\boxed{x_i=\frac{\det\mathbf{A}_i(\mathbf{b})}{\det\mathbf{A}},\qquad i=1,\dots,n.}$$

Example 50.1. The system of Part 1 §1.5, $\begin{bmatrix}2&1\\1&3\end{bmatrix}\mathbf{x}=\begin{bmatrix}5\\5\end{bmatrix}$, has $\det\mathbf{A}=5$ and

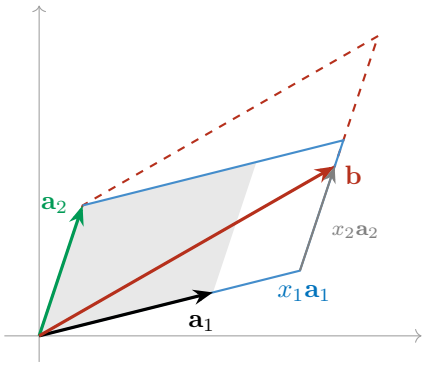
$$x_1=\frac{1}{5}\det\begin{bmatrix}5&1\\5&3\end{bmatrix}=\frac{15-5}{5}=2,\qquad x_2=\frac{1}{5}\det\begin{bmatrix}2&5\\1&5\end{bmatrix}=\frac{10-5}{5}=1.$$

50.2. Why it should be true: areas

In the plane the rule is a statement about parallelograms. The solution says $\mathbf{b}=x_1\mathbf{a}_1+x_2\mathbf{a}_2$. Compare two parallelograms that share the side $\mathbf{a}_2$: the one on $\mathbf{a}_1,\mathbf{a}_2$, of area $|\det\mathbf{A}|$ (§23), and the one on $\mathbf{b},\mathbf{a}_2$, of area $|\det\mathbf{A}_1(\mathbf{b})|$.

- Sliding $\mathbf{b}$ parallel to $\mathbf{a}_2$, back to $x_1\mathbf{a}_1$, is a shear of the parallelogram on $\mathbf{b},\mathbf{a}_2$. It keeps the base $\mathbf{a}_2$ and the height, so it keeps the area (§18.3, §22 P7).
- The parallelogram on $x_1\mathbf{a}_1,\mathbf{a}_2$ is the one on $\mathbf{a}_1,\mathbf{a}_2$ stretched by x_1 in one direction, so its area is $|x_1|$ times as large.

So $\operatorname{area}(\mathbf{b},\mathbf{a}_2)=|x_1|\cdot\operatorname{area}(\mathbf{a}_1,\mathbf{a}_2)$, and with signs, $x_1=\det[\,\mathbf{b}\ \mathbf{a}_2\,]/\det[\,\mathbf{a}_1\ \mathbf{a}_2\,]$. The unknown x_1 is a ratio of areas.



The grey parallelogram is spanned by $\mathbf{a}_1, \mathbf{a}_2$; the blue by $x_1\mathbf{a}_1, \mathbf{a}_2$; the dashed red by $\mathbf{b}, \mathbf{a}_2$. Blue and red have the same base and height, and blue is x_1 grey parallelograms long.

50.3. The proof

The picture translates into the column laws of the determinant (D6 and D7 of §38, in their column form).

Proof of Theorem 50.1. Write $\mathbf{b} = \mathbf{A}\mathbf{x} = \sum_j x_j \mathbf{a}_j$ and put it in column i:

$$\det \mathbf{A}_i(\mathbf{b}) = \det[\,\mathbf{a}_1 \;\cdots\; \textstyle\sum_j x_j\mathbf{a}_j \;\cdots\; \mathbf{a}_n\,] = \sum_j x_j \det[\,\mathbf{a}_1 \;\cdots\; \mathbf{a}_j \;\cdots\; \mathbf{a}_n\,],$$

by linearity in column i, where $\mathbf{a}_j$ now sits in position i. For $j \neq i$ that matrix has $\mathbf{a}_j$ in two columns, so its determinant is 0. Only $j = i$ survives: $\det \mathbf{A}_i(\mathbf{b}) = x_i \det \mathbf{A}$. □

In $\mathbb{R}^3$ the same argument reads: x_i is a ratio of volumes.

50.4. A formula, not an algorithm

Example 50.2. Part 1 §1.6's system, with $\det \mathbf{A} = 3$ (§42.6):

$$x_1 = \tfrac{1}{3}\det\begin{bmatrix}5&2&0\\5&1&1\\4&0&1\end{bmatrix} = \tfrac{1}{3}(5-2) = 1,$$

$$x_2 = \tfrac{1}{3}\det\begin{bmatrix}1&5&0\\0&5&1\\1&4&1\end{bmatrix} = \tfrac{1}{3}(1+5) = 2,$$

$$x_3 = \tfrac{1}{3}\det\begin{bmatrix}1&2&5\\0&1&5\\1&0&4\end{bmatrix} = \tfrac{1}{3}(4+10-5) = 3.$$

Four 3×3 determinants, where one elimination did the whole job in §42.2. In general Cramer's rule needs $n+1$ determinants of size n. That is hopeless by cofactors and still $n+1$ times the cost of elimination done sensibly. Its value is that it gives each unknown as an explicit formula in the entries. That makes it the right tool for proofs, and it produces the inverse formula of §52.

51. The inverse of an $n \times n$ matrix

51.1. The definition

Definition 51.1 (Inverse)**.** $\mathbf{A} \in \mathbb{R}^{n\times n}$ is **invertible** (or **nonsingular**) if there is a matrix $\mathbf{B} \in \mathbb{R}^{n\times n}$ with

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}.$$

Then $\mathbf{B}$ is the **inverse** of $\mathbf{A}$, written $\mathbf{A}^{-1}$. A matrix that is not invertible is **singular**.

An inverse is unique, by the argument of §34.1: $\mathbf{B} = \mathbf{B}(\mathbf{AC}) = (\mathbf{BA})\mathbf{C} = \mathbf{C}$. Only square matrices are considered. A rectangular matrix cannot have a two-sided inverse, and §54 builds the best substitute.

51.2. When a matrix is invertible

Every part of the unit so far has contributed a condition, and they are all the same condition.

Theorem 51.1 (The invertible matrix theorem)**.** *For $\mathbf{A} \in \mathbb{R}^{n\times n}$ the following are equivalent:*

(a) $\mathbf{A}$ *is invertible;*

(b) $\det \mathbf{A} \neq 0$*;*

(c) $\operatorname{rank} \mathbf{A} = n$*;*

(d) $\operatorname{null}[\mathbf{A}] = \{\mathbf{0}\}$*;*

(e) the columns of $\mathbf{A}$ *are linearly independent;*

(f) $\operatorname{col}[\mathbf{A}] = \mathbb{R}^n$*;*

(g) $\mathbf{Ax} = \mathbf{b}$ *has exactly one solution for every* $\mathbf{b} \in \mathbb{R}^n$*;*

(h) elimination on $\mathbf{A}$ *produces* n *pivots.*

Proof. (c), (d), (e), (f) are equivalent by Theorem 31.2 and Corollary 32.2 of Part 6. (b), (c), (h) are equivalent by §42.6 and §43.4. (g) is (d) and (f) together (§31.6). (a) gives (d): if $\mathbf{Ax} = \mathbf{0}$ then $\mathbf{x} = \mathbf{A}^{-1}\mathbf{Ax} = \mathbf{0}$. And (f) gives (a): this is Theorem 51.2 below. □

51.3. One side is enough

The definition asks for two equations, $\mathbf{AB} = \mathbf{I}$ and $\mathbf{BA} = \mathbf{I}$. Matrix multiplication does not commute, so it is not obvious that either one implies the other. For square matrices it does.

Theorem 51.2. *Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$. If $\mathbf{BA} = \mathbf{I}$, then also $\mathbf{AB} = \mathbf{I}$, so $\mathbf{B} = \mathbf{A}^{-1}$.*

Proof. If $\mathbf{Ax} = \mathbf{0}$ then $\mathbf{x} = \mathbf{BAx} = \mathbf{0}$, so $\operatorname{null}[\mathbf{A}] = \{\mathbf{0}\}$. By rank–nullity, $\operatorname{rank} \mathbf{A} = n$, and by Corollary 32.2, $\operatorname{col}[\mathbf{A}] = \mathbb{R}^n$. So each system $\mathbf{Ac}_j = \mathbf{e}_j$ has a solution, and the matrix $\mathbf{C} = [\,\mathbf{c}_1 \;\cdots\; \mathbf{c}_n\,]$ satisfies $\mathbf{AC} = \mathbf{I}$. Then $\mathbf{B} = \mathbf{BI} = \mathbf{B}(\mathbf{AC}) = (\mathbf{BA})\mathbf{C} = \mathbf{C}$, so $\mathbf{AB} = \mathbf{AC} = \mathbf{I}$. □

Applying the theorem with the roles of $\mathbf{A}$ and $\mathbf{B}$ exchanged, $\mathbf{AB} = \mathbf{I}$ also implies $\mathbf{BA} = \mathbf{I}$. To check that $\mathbf{B}$ is the inverse of a square matrix, one product is enough.

Orthogonal matrices, settled. §20.6 defined an orthogonal matrix by $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{QQ}^\mathsf{T} = \mathbf{I}$ and promised that the first equation implies the second. Theorem 51.2 with $\mathbf{B} = \mathbf{Q}^\mathsf{T}$ is that promise kept: *a square matrix with orthonormal columns automatically has orthonormal rows*, and

$$\mathbf{Q}^{-1} = \mathbf{Q}^\mathsf{T}.$$

Inverting an orthogonal matrix costs nothing: transpose it.

A common confusion (Square matrices only). The theorem fails for rectangular matrices. $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ has orthonormal columns and $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}_2$, but $\mathbf{A}\mathbf{A}^{\mathsf{T}} = \operatorname{diag}(1,1,0) \neq \mathbf{I}_3$.

51.4. Computing the inverse by elimination

Finding $\mathbf{A}^{-1}$ means solving the n systems $\mathbf{A}\mathbf{c}_j = \mathbf{e}_j$. They all have the same matrix, so do them at once: reduce $[\,\mathbf{A} \mid \mathbf{I}\,]$ by Gauss–Jordan elimination (§43). If $\mathbf{A}$ is invertible, the left block becomes $\mathbf{I}$ and the right block becomes $\mathbf{A}^{-1}$:

$$[\,\mathbf{A} \mid \mathbf{I}\,] \;\longrightarrow\; [\,\mathbf{I} \mid \mathbf{A}^{-1}\,].$$

If $\mathbf{A}$ is singular, a row of zeros appears on the left, and the elimination reports that no inverse exists.

Example 51.1. $\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, the matrix of Part 1 §1.6:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 + 2R_2]{R_3 \leftarrow R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 3 & -1 & 2 & 1 \end{array}\right]$$

$$\xrightarrow[R_2 \leftarrow R_2 - R_3]{R_3 \leftarrow \frac{1}{3}R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \\ 0 & 1 & 0 & \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \end{array}\right].$$

So $\mathbf{A}^{-1} = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{bmatrix}$, and $\mathbf{A}^{-1}[\,5,5,4\,] = \frac{1}{3}[\,5-10+8,\ 5+5-4,\ -5+10+4\,] = [\,1,2,3\,]$ once more.

52. The inverse by the adjugate

52.1. The adjugate

Definition 52.1 (Adjugate)**.** Let $\mathbf{C} = [\,C_{ij}\,]$ be the matrix of cofactors of $\mathbf{A} \in \mathbb{R}^{n\times n}$ (§35.2). The **adjugate** of $\mathbf{A}$ is its transpose:

$$\operatorname{adj}\mathbf{A} = \mathbf{C}^{\mathsf{T}}, \qquad (\operatorname{adj}\mathbf{A})_{ij} = C_{ji}.$$

A common confusion (Adjugate, adjoint). Older books call $\operatorname{adj}\mathbf{A}$ the *classical adjoint*, or just the *adjoint*. Most modern texts reserve "adjoint" for something else, the transpose (or, for complex matrices, the conjugate transpose). The matrix of cofactors transposed is the adjugate, whatever it is called.

52.2. The key identity

Theorem 52.1. *For every* $\mathbf{A} \in \mathbb{R}^{n\times n}$,

$$\mathbf{A}\,\operatorname{adj}\mathbf{A} = \operatorname{adj}\mathbf{A}\;\mathbf{A} = \det(\mathbf{A})\,\mathbf{I}.$$

Proof. Entry (i,k) of $\mathbf{A}\,\operatorname{adj}\mathbf{A}$ is $\sum_j a_{ij}(\operatorname{adj}\mathbf{A})_{jk} = \sum_j a_{ij}C_{kj}$.

If $i = k$, this is $\sum_j a_{ij}C_{ij}$, the Laplace expansion along row i (Theorem 36.1), which is $\det\mathbf{A}$.

If $i \neq k$, it is the expansion along row k of a different matrix: $\mathbf{A}$ with row k replaced by a copy of row i. (The cofactors C_{kj} never use row k, so they are the same for both matrices.) That matrix has two equal rows, so its determinant is 0 by D6.

So $\mathbf{A}\,\operatorname{adj}\mathbf{A}$ has $\det\mathbf{A}$ down the diagonal and 0 elsewhere. The product in the other order is the same argument with columns. □

Corollary 52.2. *If* $\det\mathbf{A} \neq 0$, *then*

$$\boxed{\mathbf{A}^{-1} = \frac{1}{\det\mathbf{A}}\,\operatorname{adj}\mathbf{A}.}$$

52.3. Examples

Example 52.1 (2×2)**.** For $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$: $C_{11} = d$, $C_{12} = -c$, $C_{21} = -b$, $C_{22} = a$. So $\mathbf{C} = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$, $\operatorname{adj}\mathbf{A} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, and the corollary is the formula of §34.2.

Example 52.2 (3×3)**.** For the matrix of Example 51.1, with $\det\mathbf{A} = 3$:

$$\begin{aligned}
C_{11} &= +\det\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = 1, & C_{12} &= -\det\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = 1, & C_{13} &= +\det\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1, \\
C_{21} &= -\det\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = -2, & C_{22} &= +\det\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = 1, & C_{23} &= -\det\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} = 2, \\
C_{31} &= +\det\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = 2, & C_{32} &= -\det\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -1, & C_{33} &= +\det\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = 1.
\end{aligned}$$

Transposing the cofactor matrix,

$$\operatorname{adj}\mathbf{A} = \begin{bmatrix} 1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{bmatrix}, \qquad \mathbf{A}^{-1} = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 2 & 1 \end{bmatrix},$$

in agreement with the elimination of Example 51.1.

52.4. It is Cramer's rule

Column j of $\mathbf{A}^{-1}$ solves $\mathbf{A}\mathbf{x} = \mathbf{e}_j$. By Cramer's rule its i-th entry is $\det\mathbf{A}_i(\mathbf{e}_j)/\det\mathbf{A}$. Expanding $\det\mathbf{A}_i(\mathbf{e}_j)$ down column i, which is $\mathbf{e}_j$, leaves the single term $1\cdot C_{ji}$. So $(\mathbf{A}^{-1})_{ij} = C_{ji}/\det\mathbf{A}$: the adjugate formula is Cramer's rule applied to each column of $\mathbf{I}$.

52.5. What it is good for

Like Cramer's rule, the adjugate is a formula, not an algorithm: it needs n^2 cofactors, each a determinant of size $n-1$. What it shows is valuable, though. Every entry of $\mathbf{A}^{-1}$ is a ratio of determinants of $\mathbf{A}$'s entries, so the inverse changes continuously as the entries of $\mathbf{A}$ do. And if $\mathbf{A}$ has integer entries and $\det \mathbf{A} = \pm 1$, then $\mathbf{A}^{-1}$ has integer entries too.

53. Properties of the inverse

53.1. The laws

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$ be invertible and $\alpha \neq 0$. By Theorem 51.2, each law is proved by checking one product.

	Law	One-line proof
I1	$(\mathbf{A}^{-1})^{-1} = \mathbf{A}$	$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$, read the other way
I2	$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$	$(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$
I3	$(\mathbf{A}^{\mathsf{T}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{T}}$	$\mathbf{A}^{\mathsf{T}}(\mathbf{A}^{-1})^{\mathsf{T}} = (\mathbf{A}^{-1}\mathbf{A})^{\mathsf{T}} = \mathbf{I}$ by M6
I4	$(\alpha\mathbf{A})^{-1} = \alpha^{-1}\mathbf{A}^{-1}$	$(\alpha\mathbf{A})(\alpha^{-1}\mathbf{A}^{-1}) = \mathbf{I}$
I5	$\det(\mathbf{A}^{-1}) = 1/\det \mathbf{A}$	$\det \mathbf{A} \det \mathbf{A}^{-1} = \det \mathbf{I} = 1$ by D9
I6	$(\mathbf{A}^k)^{-1} = (\mathbf{A}^{-1})^k$	I2, repeated

I2 reverses the order, exactly as M6 does for transposes. "First $\mathbf{B}$, then $\mathbf{A}$" is undone by "first undo $\mathbf{A}$, then undo $\mathbf{B}$". You put on socks and then shoes; you take off shoes and then socks. **I3** says the inverse of a symmetric matrix is symmetric. **I5** says an inverse undoes the volume factor.

A common confusion (No law for sums). $(\mathbf{A}+\mathbf{B})^{-1}$ is not $\mathbf{A}^{-1}+\mathbf{B}^{-1}$. Even $(\mathbf{I}+\mathbf{I})^{-1} = \frac{1}{2}\mathbf{I}$, not $2\mathbf{I}$. And $\mathbf{A}+\mathbf{B}$ can be singular when both are invertible: $\mathbf{I} + (-\mathbf{I}) = \mathbf{0}$.

53.2. Special matrices

- **Diagonal:** $\operatorname{diag}(d_1, \dots, d_n)^{-1} = \operatorname{diag}(1/d_1, \dots, 1/d_n)$, when every $d_i \neq 0$. Undo each stretch separately.
- **Triangular:** the inverse of an invertible upper triangular matrix is upper triangular, and similarly for lower. Gauss–Jordan on $[\,\mathbf{U} \mid \mathbf{I}\,]$ never needs to create an entry below the diagonal. For example, $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}^{-1} = \begin{bmatrix} 1/a & -b/(ad) \\ 0 & 1/d \end{bmatrix}$.
- **Orthogonal:** $\mathbf{Q}^{-1} = \mathbf{Q}^{\mathsf{T}}$ (§51.3). Rotations: $\mathbf{R}_\theta^{-1} = \mathbf{R}_{-\theta}$. Reflections: $\mathbf{F}^{-1} = \mathbf{F}$. A permutation matrix (a rearrangement of the rows of $\mathbf{I}$) has orthonormal columns, so its inverse is its transpose.
- **Projections:** a projection $\mathbf{P} \neq \mathbf{I}$ has $\det \mathbf{P} = 0$ (§39.4) and no inverse.

53.3. Do not invert to solve

$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ is the right way to *write* the solution, and almost never the right way to *compute* it. Computing $\mathbf{A}^{-1}$ costs about three times as much as one elimination, and the extra arithmetic adds

rounding error. To solve $\mathbf{Ax} = \mathbf{b}$, eliminate (Part 8). Keep $\mathbf{A}^{-1}$ as notation, for formulas and for proofs.

53.4. Least squares, with an inverse

When $\mathbf{A}$ has full column rank, $\mathbf{A}^\mathsf{T}\mathbf{A}$ has rank n (§33.4), so it is invertible (Theorem 51.1), and the normal equations of Part 9 can be solved in one line:

$$\hat{\mathbf{x}} = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}\mathbf{b}.$$

For the line fit of §49.4, $\mathbf{A}^\mathsf{T}\mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$ has determinant 6, and

$$\hat{\mathbf{x}} = \frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}\begin{bmatrix} 5 \\ 11 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix},$$

the same line $y = \frac{2}{3} + \frac{1}{2}t$.

54. The pseudo-inverse

54.1. An inverse for every matrix

Rectangular and singular matrices have no inverse. Yet §53.4 just produced, for a tall matrix with independent columns, a matrix that turns $\mathbf{b}$ into the best available $\mathbf{x}$: the matrix $(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}$ in front of $\mathbf{b}$. When $\mathbf{A}$ is square and invertible, the best $\mathbf{x}$ is the exact solution $\mathbf{A}^{-1}\mathbf{b}$. So this matrix behaves like an inverse wherever an inverse exists, and like the best substitute where one does not. That is the idea of the pseudo-inverse.

54.2. Full column rank

Definition 54.1 (Pseudo-inverse, full column rank)**.** If $\mathbf{A} \in \mathbb{R}^{m\times n}$ has independent columns, its **pseudo-inverse** is the $n \times m$ matrix

$$\mathbf{A}^+ = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}.$$

Then $\mathbf{A}^+\mathbf{b}$ is the least-squares solution, and:

- $\mathbf{A}^+\mathbf{A} = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}\mathbf{A} = \mathbf{I}_n$: it is a *left* inverse.
- $\mathbf{AA}^+ = \mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}$ is $m \times m$. It is symmetric (by I3 and M6) and idempotent: $\big(\mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}\big)^2 = \mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}(\mathbf{A}^\mathsf{T}\mathbf{A})(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T} = \mathbf{AA}^+$. So it is an orthogonal projection (§27.4), and it sends $\mathbf{b}$ to $\mathbf{A}\hat{\mathbf{x}}$, the closest point of col[$\mathbf{A}$]. It is *the projection matrix onto the column space*. Unless $m = n$, it is not $\mathbf{I}$, so $\mathbf{A}^+$ is not a right inverse.

Example 54.1 (One column)**.** For a single nonzero column $\mathbf{u} \in \mathbb{R}^m$, $\mathbf{u}^\mathsf{T}\mathbf{u}$ is a number and

$$\mathbf{u}^+ = \frac{\mathbf{u}^\mathsf{T}}{\mathbf{u}^\mathsf{T}\mathbf{u}}, \qquad \mathbf{u}\,\mathbf{u}^+ = \frac{\mathbf{u}\mathbf{u}^\mathsf{T}}{\mathbf{u}^\mathsf{T}\mathbf{u}} = \mathbf{P}_\mathbf{u}.$$

The projection matrix of §26 was $\mathbf{uu}^+$ all along, and $\mathbf{u}^+\mathbf{v} = \mathbf{u}^\mathsf{T}\mathbf{v}/\mathbf{u}^\mathsf{T}\mathbf{u}$ is the $\hat{x}$ of §46.

Example 54.2 (The line fit)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$,

$$\mathbf{A}^+ = \frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 8 & 2 & -4 \\ -3 & 0 & 3 \end{bmatrix}, \qquad \mathbf{A}\mathbf{A}^+ = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}.$$

$\mathbf{A}^+[\,1,2,2\,] = \frac{1}{6}[\,4,3\,]$, as before, and $\mathbf{A}^+\mathbf{A} = \mathbf{I}_2$. The projection $\mathbf{A}\mathbf{A}^+$ is symmetric with trace $2 = \operatorname{rank}\mathbf{A}$.

54.3. Full row rank: the shortest solution

If $\mathbf{A} \in \mathbb{R}^{m\times n}$ has independent *rows* ($m < n$), every system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent and has infinitely many solutions (§45.2). Now $\mathbf{A}\mathbf{A}^\mathsf{T}$ is $m \times m$ with rank $\operatorname{rank}\mathbf{A}^\mathsf{T} = m$ (§33), so it is invertible.

Definition 54.2 (Pseudo-inverse, full row rank)**.** If $\mathbf{A}$ has independent rows, $\mathbf{A}^+ = \mathbf{A}^\mathsf{T}(\mathbf{A}\mathbf{A}^\mathsf{T})^{-1}$.

Now $\mathbf{A}\mathbf{A}^+ = \mathbf{I}_m$, a *right* inverse, so $\mathbf{x}^+ = \mathbf{A}^+\mathbf{b}$ solves $\mathbf{A}\mathbf{x} = \mathbf{b}$ exactly. Of all the solutions, it is the shortest.

Theorem 54.1. *If* $\mathbf{A}$ *has independent rows,* $\mathbf{x}^+ = \mathbf{A}^+\mathbf{b}$ *is the solution of* $\mathbf{A}\mathbf{x} = \mathbf{b}$ *of smallest length.*

Proof. $\mathbf{x}^+ = \mathbf{A}^\mathsf{T}\big((\mathbf{A}\mathbf{A}^\mathsf{T})^{-1}\mathbf{b}\big)$ is $\mathbf{A}^\mathsf{T}$ times a vector, so it lies in $\operatorname{col}[\mathbf{A}^\mathsf{T}] = \operatorname{row}[\mathbf{A}]$. Any other solution is $\mathbf{x} = \mathbf{x}^+ + \mathbf{z}$ with $\mathbf{z} \in \operatorname{null}[\mathbf{A}]$ (§31.6), and $\operatorname{null}[\mathbf{A}] \perp \operatorname{row}[\mathbf{A}]$ (§31.7). By the Pythagorean identity, $\|\mathbf{x}\|^2 = \|\mathbf{x}^+\|^2 + \|\mathbf{z}\|^2 \geq \|\mathbf{x}^+\|^2$. □

Example 54.3. $x_1 + x_2 + x_3 = 3$ (§45.2): $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$, $\mathbf{A}\mathbf{A}^\mathsf{T} = 3$, $\mathbf{A}^+ = \frac{1}{3}[\,1,1,1\,]^\mathsf{T}$, and $\mathbf{x}^+ = [\,1,1,1\,]$, the shortest point of the plane of solutions, as §45.2 predicted.

If $\mathbf{A}$ is square and invertible, both definitions give $\mathbf{A}^{-1}$. For example, $(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T} = \mathbf{A}^{-1}(\mathbf{A}^\mathsf{T})^{-1}\mathbf{A}^\mathsf{T} = \mathbf{A}^{-1}$.

54.4. The Moore–Penrose definition

The two formulas need full rank of one kind or the other. A single definition covers every matrix.

Definition 54.3 (Moore–Penrose pseudo-inverse)**.** For any $\mathbf{A} \in \mathbb{R}^{m\times n}$, the **pseudo-inverse** $\mathbf{A}^+ \in \mathbb{R}^{n\times m}$ is the matrix $\mathbf{X}$ satisfying the four conditions

$$\textbf{(MP1)}\ \ \mathbf{A}\mathbf{X}\mathbf{A} = \mathbf{A}, \qquad \textbf{(MP2)}\ \ \mathbf{X}\mathbf{A}\mathbf{X} = \mathbf{X},$$
$$\textbf{(MP3)}\ \ (\mathbf{A}\mathbf{X})^\mathsf{T} = \mathbf{A}\mathbf{X}, \qquad \textbf{(MP4)}\ \ (\mathbf{X}\mathbf{A})^\mathsf{T} = \mathbf{X}\mathbf{A}.$$

It is a theorem, stated here without proof, that *every* matrix has exactly one such $\mathbf{X}$. The general construction uses the singular value decomposition, at the end of the course. The conditions say that $\mathbf{A}\mathbf{A}^+$ and $\mathbf{A}^+\mathbf{A}$ are orthogonal projections (MP1–MP4 make them symmetric and idempotent), onto $\operatorname{col}[\mathbf{A}]$ and onto $\operatorname{row}[\mathbf{A}]$. In general $\mathbf{A}^+\mathbf{b}$ is the *shortest* of the least-squares solutions.

Checking the full-column-rank formula. With $\mathbf{X} = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}$ we have $\mathbf{X}\mathbf{A} = \mathbf{I}$, which is symmetric (MP4). Then $\mathbf{A}(\mathbf{X}\mathbf{A}) = \mathbf{A}$ (MP1) and $(\mathbf{X}\mathbf{A})\mathbf{X} = \mathbf{X}$ (MP2). And $\mathbf{A}\mathbf{X}$ is symmetric, as shown in §54.2 (MP3). The full-row-rank formula checks the same way, with $\mathbf{A}\mathbf{X} = \mathbf{I}$.

54.5. A rank-deficient example

Example 54.4. A rank-one matrix $\mathbf{A} = \mathbf{x}\mathbf{y}^{\mathsf{T}}$ (with $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{y} \in \mathbb{R}^n$ nonzero) has neither independent columns nor independent rows when $m, n \geq 2$, so neither formula applies. The pseudo-inverse is

$$\mathbf{A}^{+} = \frac{\mathbf{y}\mathbf{x}^{\mathsf{T}}}{\|\mathbf{x}\|^2 \|\mathbf{y}\|^2}.$$

Check, with $\mathbf{x}^{\mathsf{T}}\mathbf{x} = \|\mathbf{x}\|^2$ and $\mathbf{y}^{\mathsf{T}}\mathbf{y} = \|\mathbf{y}\|^2$:

$$\mathbf{A}\mathbf{A}^{+} = \frac{\mathbf{x}(\mathbf{y}^{\mathsf{T}}\mathbf{y})\mathbf{x}^{\mathsf{T}}}{\|\mathbf{x}\|^2\|\mathbf{y}\|^2} = \frac{\mathbf{x}\mathbf{x}^{\mathsf{T}}}{\|\mathbf{x}\|^2} = \mathbf{P}_{\mathbf{x}}, \qquad \mathbf{A}^{+}\mathbf{A} = \frac{\mathbf{y}(\mathbf{x}^{\mathsf{T}}\mathbf{x})\mathbf{y}^{\mathsf{T}}}{\|\mathbf{x}\|^2\|\mathbf{y}\|^2} = \mathbf{P}_{\mathbf{y}},$$

both symmetric (MP3, MP4). Then $\mathbf{A}\mathbf{A}^{+}\mathbf{A} = \mathbf{P}_{\mathbf{x}}\mathbf{x}\mathbf{y}^{\mathsf{T}} = \mathbf{x}\mathbf{y}^{\mathsf{T}} = \mathbf{A}$ (MP1), and $\mathbf{A}^{+}\mathbf{A}\mathbf{A}^{+} = \mathbf{P}_{\mathbf{y}}\mathbf{y}\mathbf{x}^{\mathsf{T}}/(\cdots) = \mathbf{A}^{+}$ (MP2). For instance $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = [\,1,2\,]^{\mathsf{T}}[\,1,2\,]$ has pseudo-inverse $\frac{1}{25}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, although it has no inverse.

54.6. Practice

P1. Solve $2x + y = 5$, $x - y = 1$ by Cramer's rule.

P2. Find the inverse of $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ by Gauss–Jordan elimination.

P3. Find the inverse of $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ by the adjugate.

P4. $\mathbf{A}, \mathbf{B}$ are 3×3 with $\det \mathbf{A} = 2$, $\det \mathbf{B} = 3$. Find $\det\big((\mathbf{A}\mathbf{B})^{-1}\big)$, $\det(\mathbf{A}^{-1}\mathbf{B}^{\mathsf{T}})$ and $\det(2\mathbf{A}^{-1})$.

P5. Prove that if $\mathbf{A}$ is invertible and symmetric, so is $\mathbf{A}^{-1}$.

P6. Find the pseudo-inverse of the column $\mathbf{u} = [\,1,2,2\,]$, and compute $\mathbf{u}^{+}\mathbf{u}$ and $\mathbf{u}\mathbf{u}^{+}$.

P7. Find the shortest solution of $x_1 + 2x_2 = 5$.

Answers

P1. $\det = -2 - 1 = -3$; $x = \det\begin{bmatrix} 5 & 1 \\ 1 & -1 \end{bmatrix}/(-3) = (-6)/(-3) = 2$, $y = \det\begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix}/(-3) = (-3)/(-3) = 1$.

P2. The top-left block $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ has determinant 1 and inverse $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$, and the 3 inverts to $\frac{1}{3}$: $\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$. (Elimination never mixes the third row with the first two.)

P3. $\det = -1$ (expand along row 2). Cofactors: row 1: $1, 0, -1$; row 2: $0, -1, 0$; row 3: $-2, 0, 1$. So $\operatorname{adj}\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ and $\mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 2 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$. Check: row 1 of $\mathbf{A}$ times the columns gives $[\,-1+2,\ 0,\ 2-2\,] = [\,1,0,0\,]$. ✓

P4. $\frac{1}{6}$; $\frac{3}{2}$; $2^3 \cdot \frac{1}{2} = 4$.

P5. By I3, $(\mathbf{A}^{-1})^{\mathsf{T}} = (\mathbf{A}^{\mathsf{T}})^{-1} = \mathbf{A}^{-1}$.

P6. $\mathbf{u}^{+} = \frac{1}{9}[1, 2, 2]$ (a row), $\mathbf{u}^{+}\mathbf{u} = 1$, and $\mathbf{u}\mathbf{u}^{+} = \frac{1}{9}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix} = \mathbf{P}_{\mathbf{u}}$.

P7. $\mathbf{A} = [\,1 \;\; 2\,]$, $\mathbf{A}\mathbf{A}^{\mathsf{T}} = 5$, so $\mathbf{x}^{+} = \frac{5}{5}[1, 2] = [1, 2]$, of length $\sqrt{5}$; and $1 + 4 = 5$. ✓

Looking ahead. The pseudo-inverse of a matrix with independent columns was $(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}$, and forming $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ is exactly the step that numerical analysts avoid. Part 11 builds an orthonormal basis for the column space with the Gram–Schmidt process and records it as a factorization $\mathbf{A} = \mathbf{Q}\mathbf{R}$. With it, least squares becomes a single back substitution, and the pseudo-inverse becomes $\mathbf{R}^{-1}\mathbf{Q}^{\mathsf{T}}$.